TripNest – The Traveller Insights Challenge

Current Scenario

TripNest is a digital travel platform that helps travellers discover and book accommodations, activities, and experiences. It collects data from four main sources: travellers, property partners, experience providers, and customer support. This data is used by the Travel, Partner, and Business teams to understand traveller preferences, manage partner information, and analyze bookings and customer behaviour. To support consistent and reliable decision-making, TripNest needs to organize data from these sources into accurate, standardized, and trustworthy information.

Data from Four Sources

1. Travellers Data

Traveller ID	Name	Email	Phone	Preferences
TR001	Arun Kumar	arun@email.com	9876501234	Beach stays
TR002	A. Kumar	arun@email.com	9876501234	Adventure trips

2. Property Partners Data

Property ID	Partner Name	Property Name	Location	Availability
PR001	Sunrise Hotels	Sunrise Resort	Goa	Available
PR002	CityStay Group	CityStay Hotel	Bengaluru	Unavailable

3. Experience Providers Data

Provider ID	Provider Name	Experience Name	Location	Availability
EP001	Ocean Adventures	Scuba Diving	Goa	Available
EP002	Hilltop Tours	Mountain Trekking	Coorg	Available

4. Customer Support Data

Support Case ID	Traveller ID	Booking ID	Issue	Resolution Status
CS001	TR001	BK101	Booking cancellation	Resolved
CS002	TR002	BK102	Payment issue	Pending

Problems:

1. Duplicate Traveller Records

- The same traveller may appear under different names or contact details, and may use different devices, making it difficult to identify whether the records belong to the same person.
- **Issue: Entity Duplication** - Identifying and merging records belonging to the same traveller is difficult.

2. Inconsistent Business Definitions

- Different teams may define a “completed booking” differently: one team may count successful payment, while another counts only when the traveller uses the service.
- **Issue: Metric Inconsistency** - Teams calculate business metrics using different definitions.

3. Changing Booking Information

- Booking information may change after reporting, such as when a confirmed booking is later cancelled by a property partner.
- **Issue: Data Freshness** - Reports may contain outdated booking statuses.

4. Data from Multiple Sources

- TripNest collects data from travellers, property partners, experience providers, and customer support, with different formats and structures.
- **Issue: Data Integration** - Combining data from different sources into a consistent format.

5. Different Team Requirements

- The Travel Team needs traveller preferences, the Partner Team needs property and experience details, and the Business Team needs booking and customer behaviour information.
- **Issue: Data Organization -** Data must be structured to meet each team's requirements consistently.

6. Lack of Data Traceability

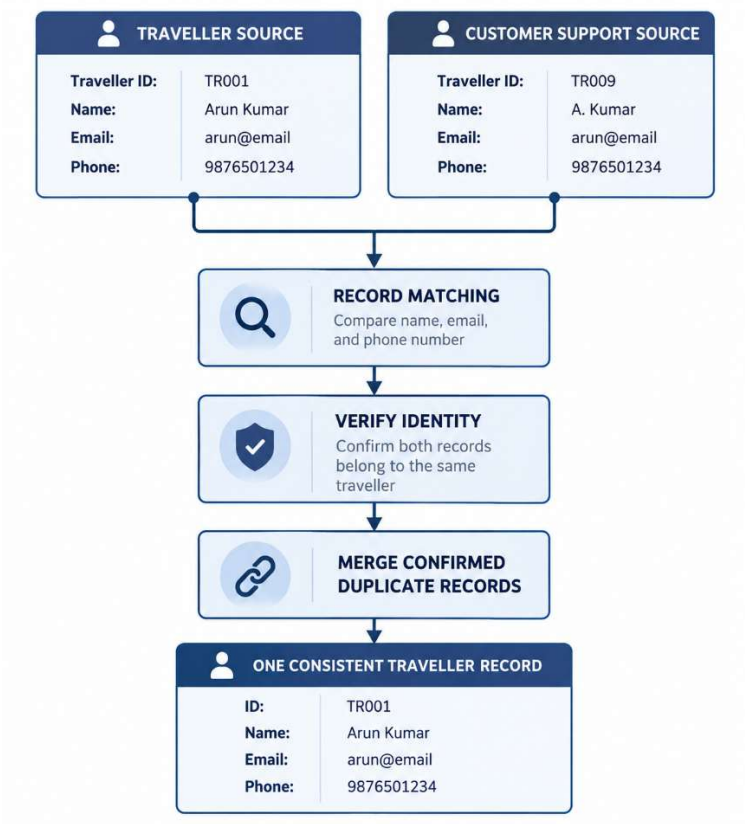
- TripNest needs to identify where information in a report originated and what changes were made before the information reached that report.
- **Issue: Data Lineage -** Tracking the source and transformation history of data is difficult.

Solutions:

Problem 1: Duplicate Traveller Records

Solution: Entity Resolution

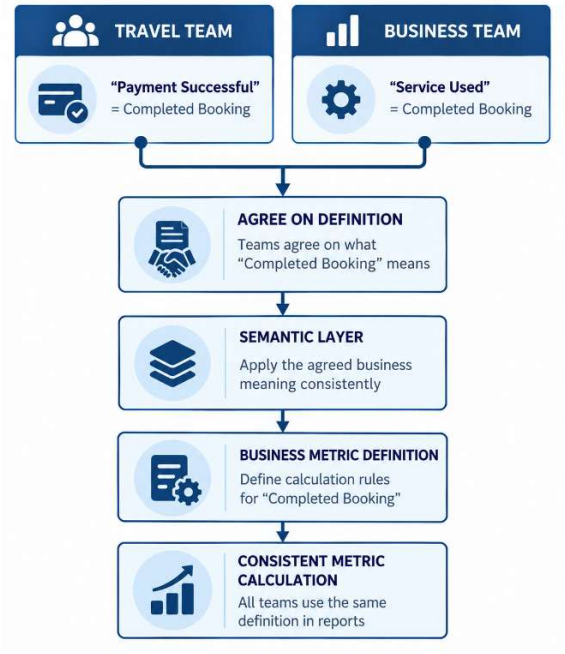
- **Record Matching:** Compare traveller details, such as names, email addresses, and phone numbers, across multiple sources to identify records that may belong to the same traveller.
- **Duplicate Resolution:** Verify the matching records and merge confirmed duplicates into one consistent traveller record while preserving source references and relevant history.



Problem 2: Inconsistent Business Definitions

Solution: Semantic Layer and Business Metric Definitions

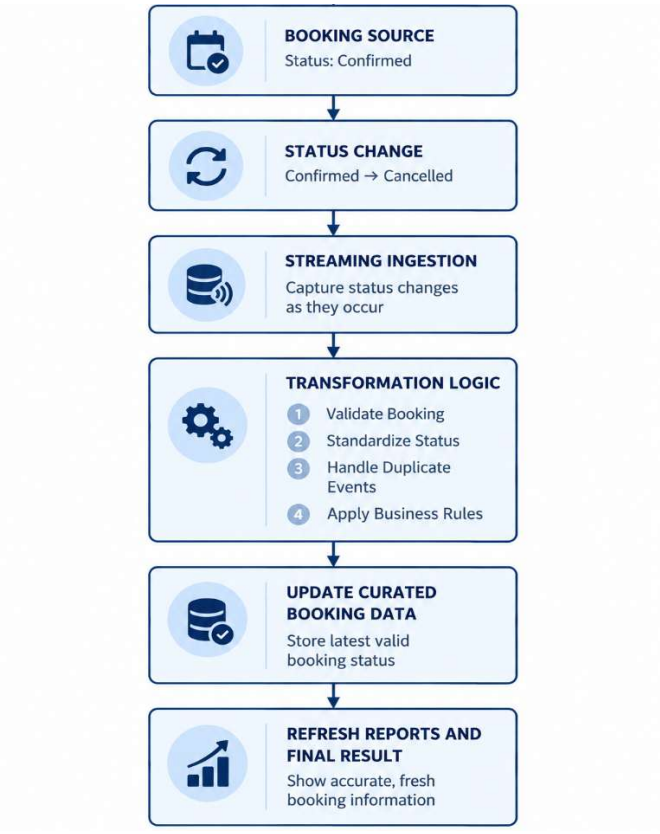
- **Standardize Business Definitions:** Define “completed booking” clearly and get agreement from all teams on whether it means successful payment or actual use of the booked service.
- **Consistent Metric Calculation:** Apply the agreed definition consistently across reports and dashboards so all teams calculate and interpret booking metrics in the same way.



Problem 3: Changing Booking Information

Solution: Streaming Ingestion, Transformation Logic, and Data Freshness Management

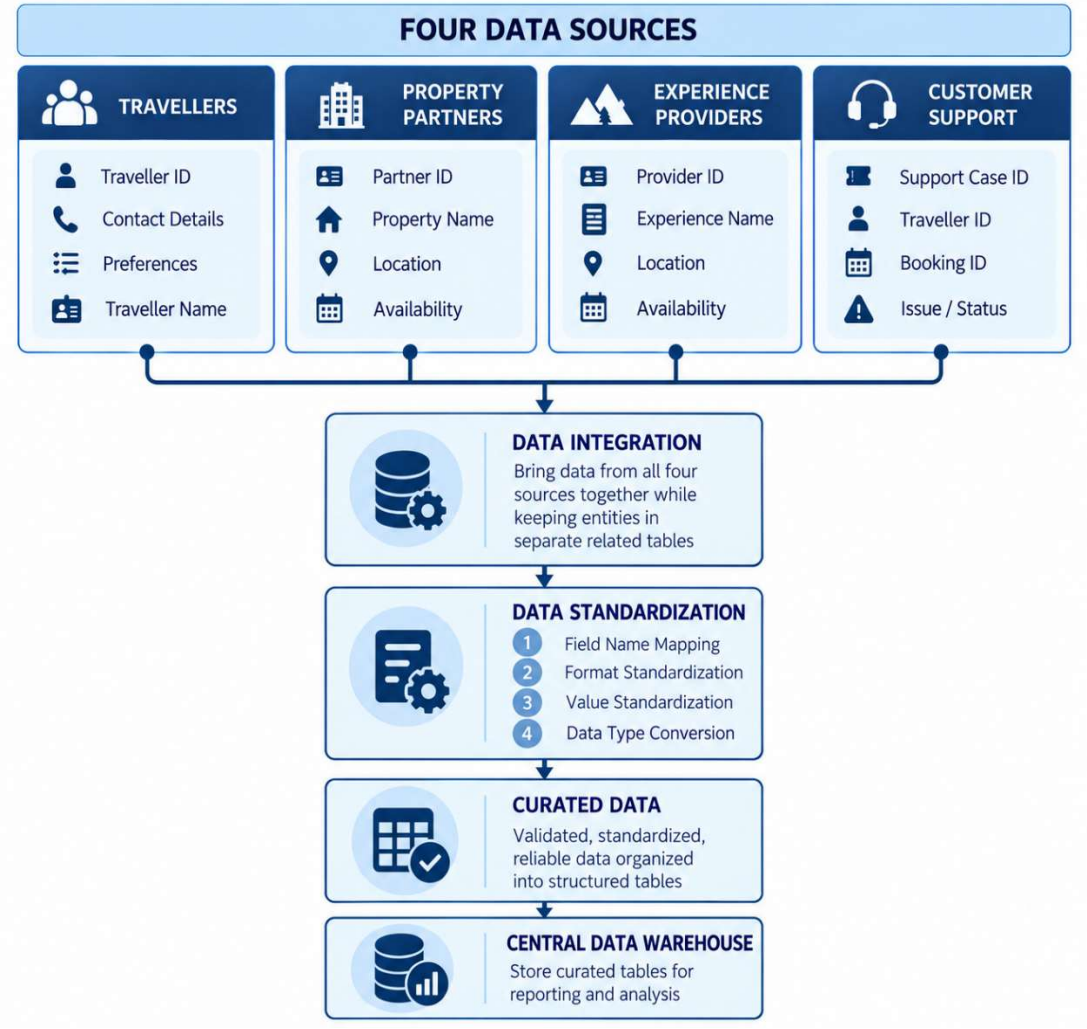
- **Streaming Ingestion and Change Processing:** Capture booking status changes, such as Confirmed → Cancelled, as they occur, then process events using transformation logic.
- **Data Freshness and Report Updates**: Update curated booking data and refresh reports promptly so teams use the latest valid booking status instead of outdated information.



Problem 4: Data from Multiple Sources

Solution: Data Integration and Standardization

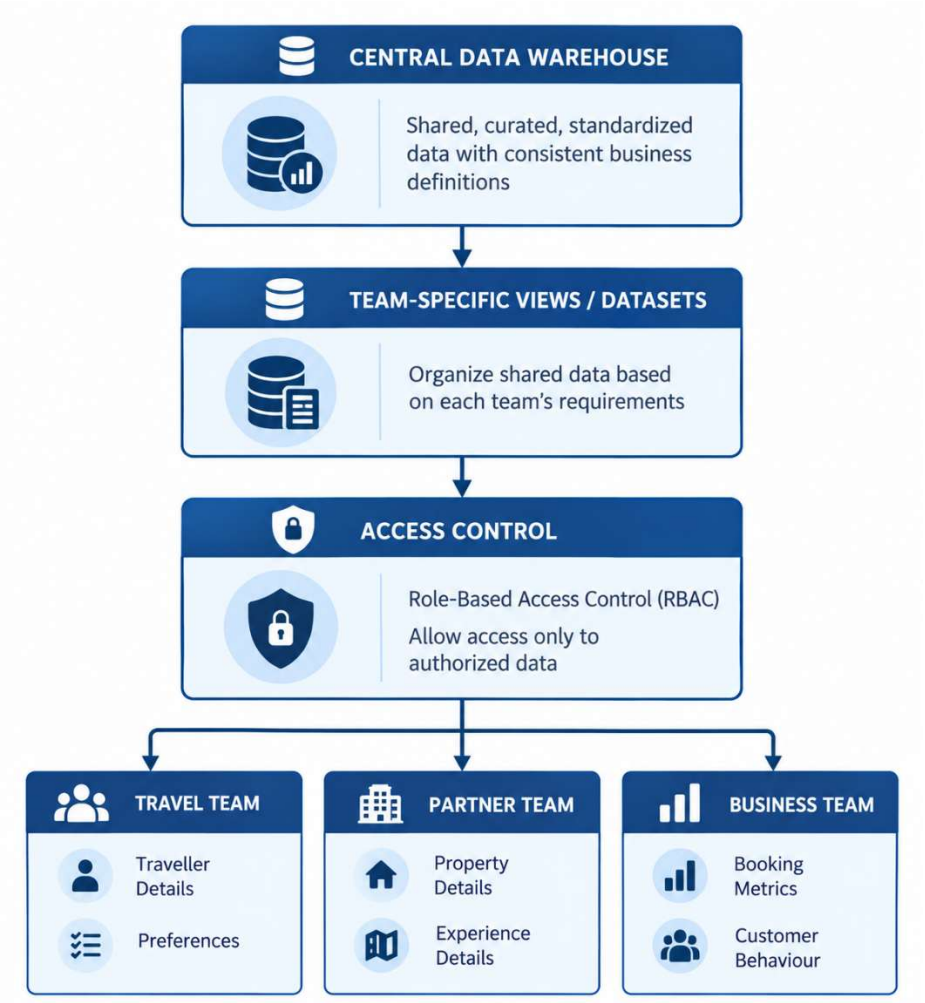
- **Data Integration:** Bring data from all four sources into a central data platform while keeping their data in separate tables and connecting related records where required.
- **Data Standardization:** Apply field name mapping, format standardization, value standardization, and data type conversion to ensure consistent data, then organize and store it in a Central Data Warehouse for reporting and analysis.



Problem 5: Different Team Requirements

Solution: Team-Specific Data Organization

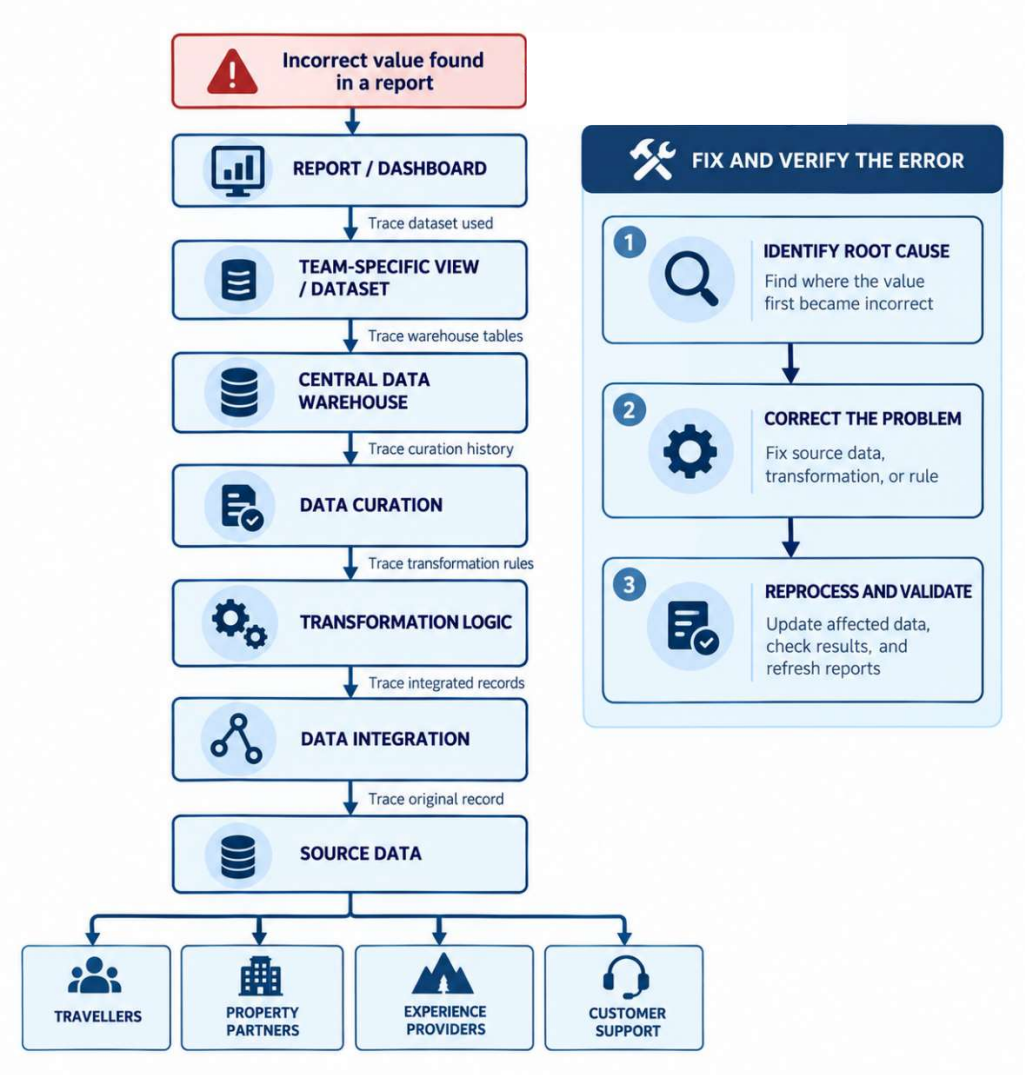
- **Create Team-Specific Views/Datasets:** Organize shared curated data into views or datasets based on each team's requirements, providing relevant information to the Travel, Partner, and Business teams.
- **Ensure Consistency and Access Control:** Use standardized data and consistent business definitions across teams, and apply Role-Based Access Control (RBAC) so each team accesses only authorized data without unnecessarily duplicating the underlying data.



Problem 6: Lack of Data Traceability

Solution: Data Lineage

- **Track Data Origin and Transformations:** Maintain data lineage to trace data from the four original sources through integration, transformation, curation, and warehouse storage to team-specific datasets and reports.
- **Identify and Fix Errors:** When incorrect data appears in a report, trace it backward through each stage to find the root cause. Correct the source data, transformation logic, or business rule, then reprocess affected data and validate the updated reports.



Proposed Architecture:

